Retinal bottleneck in anatomically constrained CNNs

A reproduction of Lindsey et al. (ICLR 2019) on STL-10 and Imagenette

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Abstract

Lindsey et al. (2019) train a convolutional network on object recognition, split into a "retina" and a "ventral visual stream", and show that two classical facts of visual neuroscience emerge from a single model: the retinal output has center-surround receptive fields while the first cortical layer has oriented ones, provided that the retinal output is a narrow bottleneck; and the depth of the cortex determines how linear and information-preserving the retina is. I reproduce their experiments on STL-10 and Imagenette (grayscale, 96×96, 4 500–8 500 training images against 45 000) with 3 seeds against 10 and a single laptop GPU. I train with two recipes: a *paper recipe*, which follows the text of the paper and fills what it does not state (above all the learning rate) with Keras defaults, and a *code recipe*, copied from the authors' released code (learning rate 10^{-4}, ten times lower than the Keras default, batch 64 instead of the 32 stated in the text, shift augmentation, black-image probe). With the paper recipe most retinal channels die even in the paper's own setting; the key configurations were therefore run with the code recipe (36 networks), under a decision rule written before running. With the code recipe the paper's own setting (CIFAR-10) is reproduced: no dead channels, a one-channel retina with an orientation index of 1.83 ± 0.38 (paper: 1.56 ± 0.22; all three networks center-surround), a first cortical layer at 3.22 ± 0.62 (paper: 3.05 ± 0.30), and an oriented 32-channel retina. On STL-10 and Imagenette, with the same recipe and the paper's number of weight updates, accuracy grows with the bottleneck width as in the paper, but the one-channel retina is oriented (index 3.42 and 3.30 under a natural-image probe; none of six channels center-surround), so the central claim does not transfer to the assignment's datasets. The runs with the paper recipe (234 networks, including a cortex-depth grid and a bridge experiment between the two settings) cover the claims that the code recipe does not.

1 Introduction

Why do retinal ganglion cells respond to local contrast while neurons in the primary visual cortex (V1) respond to oriented edges? Classical theories of efficient coding explain each geometry separately, but not why two consecutive stages should differ. Lindsey et al. [1] propose that the difference follows from a *resource constraint*: the optic nerve carries the signal of roughly a hundred million photoreceptors through about one million fibres, so the retina is a dimensionality bottleneck. They model the visual system as a deep CNN trained for object recognition, vary the width of the retinal output and the depth of the cortex, and look at the receptive fields (RFs) that emerge. The assignment of this project is a *reproduction* of that paper on two datasets chosen by the instructor, STL-10 and Imagenette at 96×96, with ablations of bottleneck width, retina depth and cortex depth, and three seeds per configuration.

The text of the paper does not state the learning rate, the augmentation, the input scaling or the checkpoint, and the authors' code differs from the text on the batch size and on the bias initialisation. I therefore compare two recipes: the *paper recipe* follows the text and fills the gaps with Keras defaults (learning rate 10^{-3}); the *code recipe* copies the authors' code (learning rate 10^{-4}). With the paper recipe the results are mixed, and a positive control in the paper's own setting fails, with most retinal channels dead. With the code recipe, for which I wrote down the differences, a prediction and a decision rule before running, I trained the paper's setting and the key configurations of the assignment. Those runs are the main results

(Section 5); the runs with the paper recipe (Section 6) cover configurations that were not repeated and show what goes wrong when the text is followed. Section 2 gives the neuroscience needed to read the paper, Section 3 summarises it, Section 4 describes the setup and every difference from the original, and Sections 7–9 discuss, list the limitations and conclude.

2 Background: the early visual system

From the eye to the cortex. Light is transduced by photoreceptors (rods and cones). Their signal passes through bipolar cells to the *retinal ganglion cells*, whose axons form the optic nerve. In humans there are of the order of 10^8 photoreceptors and 10^6 ganglion cells, a compression of about two orders of magnitude, so the optic nerve is a stringent bottleneck. The signal is relayed by the thalamus (lateral geniculate nucleus, LGN) to the primary visual cortex V1 and then along the *ventral stream* (V2, V4, inferotemporal cortex), which supports object recognition.

Receptive fields. The receptive field (RF) of a neuron is the part of the visual scene, and the kind of pattern, that drives it. Two geometries matter here (Fig. 1b).

- *Center-surround* RFs of retinal ganglion cells (and LGN): a small excitatory center surrounded by an inhibitory ring, or the opposite (ON and OFF cells). They respond to local contrast and not to uniform illumination, and act as a decorrelating (whitening) filter.
- *Oriented* RFs of V1 simple cells: elongated patches responding to an edge of a given orientation and position, well described by Gabor functions. Hubel and Wiesel [5] hypothesised that they are built by pooling center-surround inputs aligned along the edge.

The open question. Efficient-coding theories reproduce center-surround RFs (decorrelation of natural images under noise [6]) and oriented RFs (sparse coding [7]), but with different objectives and constraints for the two stages. Moreover, retinas differ across species: in primates the most common ganglion cell type is close to a linear pixel encoder, while in the mouse it is a nonlinear feature detector [8]. A single account should explain both the retina–V1 difference and the diversity across species.

3 The reference paper

Model. The paper trains a CNN on grayscale CIFAR-10 [2] (Fig. 1a). A *retina-net* of two convolutions (32 channels, then N_{BN} channels, the retinal output) feeds a *ventral-visual-system net* (VVS-net) of D_{VVS} convolutions with 32 channels followed by two dense layers (1024 and 10 units). All convolutions are 9×9, stride 1, with ReLU and no batch normalisation; the text states RMSProp for 20 epochs, batch size 32, Glorot-uniform initialisation [4] and ten random initialisations per architecture. The RF of a channel is the gradient of its activation at the image centre with respect to a blank image, equivalent to a spike-triggered average for a small perturbative stimulus.

Claims. (i) *A bottleneck at the retinal output* ($N_{\mathrm{BN}} = 4, 2, 1$) *yields center-surround retinal RFs and oriented RFs in the first cortical layer*; with $N_{\mathrm{BN}} = 32$ the retinal RFs are oriented. The RF geometry does not depend qualitatively on D_{VVS} (shown for $D_{\mathrm{VVS}} = 2$), except for $D_{\mathrm{VVS}} = 0$. (ii) *The cortex depth sets the linearity of the retina:* with a deeper cortex the retinal response is more linear and keeps more information about the image but separates the classes less; with a shallow cortex the retina is nonlinear and extracts features. (iii) *The accuracy increases with the bottleneck width and with the cortex depth* (their Fig. 1D).

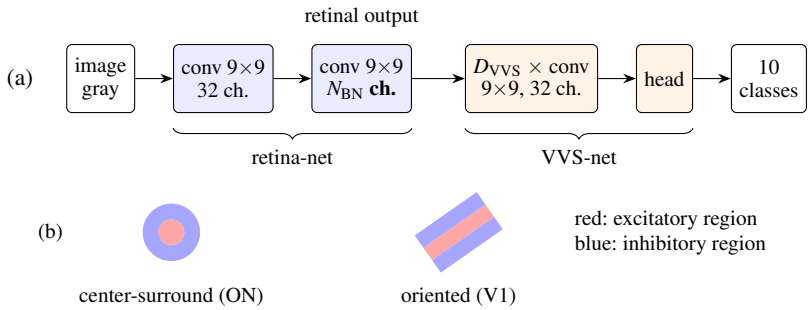


Figure 1: (a) The architecture of the paper and of this project: a retina-net whose output is a bottleneck of N_{BN} channels, followed by a cortex of D_{VVS} convolutions and a classifier head. (b) The two receptive-field geometries of the early visual system.

Measures. The *orientation selectivity index* (OSI) is computed from the responses of an RF to rectangular bars of all widths, orientations and positions: the ratio between the strongest response at the preferred orientation and the strongest response at the orthogonal one. It is ≈ 1 for isotropic RFs and larger for oriented ones. The paper reports 1.56 ± 0.22 for the retinal output ($N_{BN} = 1$, $D_{VVS} = 2$), 3.05 ± 0.30 for the first cortical layer, and 1.97 ± 0.08 for random-noise RFs. *Linearity* is the Pearson correlation between the true response of a retinal unit and its ridge-regression prediction from the image; *information* is the variance explained when the image is linearly reconstructed from the retinal output.

4 My reproduction: setup and differences

4.1 Hypotheses of the assignment

The assignment is phrased as three hypotheses: (H1) a narrow bottleneck with a deep cortex makes the retina learn center-surround filters; (H2) a wide bottleneck or a shallow cortex makes the retina learn oriented filters; (H3) the accuracy degrades little as the bottleneck narrows, until a minimum width. H1 and H2 are stricter than claims (i) and (ii): the paper does not tie center-surround RFs to a deep cortex, nor oriented retinal RFs to a shallow one, and its cortex-depth effect is on the *linearity* of the retina. I therefore test both the hypotheses as written and the paper's own claims.

4.2 Data

STL-10 and Imagenette are decoded once (resize to 96, centre crop, grayscale) and kept as 8-bit tensors on the GPU. The test sets are the official STL-10 `test` split and the Imagenette `val` split. CIFAR-10 is not simply "easier": it has far more labels per class (5 000 against 500 for STL-10) but a much lower resolution. The two properties that matter for the mechanism studied here are the resolution, relative to the

	Resolution	Classes	Labelled train / test	Role here
CIFAR-10	32×32 colour	10	50 000 / 10 000	paper's data; positive control
STL-10	96×96 colour	10	5 000 / 8 000 (+100 000 unlabelled, unused)	main dataset
Imagenette	up to 160 px	10 easy ImageNet classes	9 469 / 3 925	main dataset

Table 1: Datasets. STL-10 [3] is a subset of ImageNet with few labels per class; Imagenette is a ten-class ImageNet subset by fast.ai. 10% of each training set is held out for validation, so the networks are trained on about 4 500 (STL-10), 8 500 (Imagenette) and 45 000 (CIFAR-10) images.

	Paper: text, and code where it differs	Main runs (code recipe)	Paper recipe
Data	grayscale CIFAR-10, 32×32	CIFAR-10 control; STL-10 and Imagenette at 96×96 (assignment)	the same
Head	flatten, dense 1024, dense 10	paper's on CIFAR-10; at 96×96: 2×2 max-pool after every cortical conv, global average pool, linear layer	the same
Optimiser	RMSProp, 20 epochs, batch 32; code: lr 10^{-4} with decay 10^{-6}, batch 64	as the code; on STL-10 and Imagenette 221 and 117 epochs, the paper's number of updates (15 640)	RMSProp lr 10^{-3}, batch 32, 20 epochs
Initialisation	text: Glorot weights and biases; code: biases zero	Glorot weights, zero biases	the same
Input, augmentation	not stated; code: mean of RGB in $[0,1]$, shifts up to 10% and horizontal flip	the same (torchvision grayscale weights)	centred as $(x-0.5)/0.25$; flip only
Model kept	not stated; code: last epoch	last epoch	best validation epoch
Collapsed runs	not discussed	retrain once with lr/3 if validation accuracy < 0.2 (never needed)	the same (needed in 25 runs)
RF probe	blank image; code: all-zero image, zero-gradient RFs dropped from averages	both the black image (as the code) and 256 natural images	256 natural images
OSI	code: 32×32 RF, bars of 50 orientations, 30 widths, 30 positions, smoothed	the code, ported (noise: 1.925; paper 1.97)	own version, 17×17 crop
Seeds, compute	10 per architecture	3; about 8.5 h overnight	3 (5 in the bridge); about 10 h
Not reproduced	$D_{\mathrm{VVS}} = 0$, class separability, untied retina, LGN, simple/complex cells, local normalisation; PyTorch instead of Keras.		

Table 2: Differences between the original setup, the main runs and the runs with the paper recipe. One laptop GPU (RTX PRO 2000, 8 GB) for both.

9×9 kernels, and the amount of training data.

4.3 Architecture, training and differences from the paper

My network follows Fig. 1a. On CIFAR-10 it is the paper's architecture; on STL-10 and Imagenette the head differs, because a flatten and a dense layer of 1024 units at 96×96 would have hundreds of millions of weights. Table 2 lists every difference, for the main runs (the code recipe, taken from the authors' code[1]) and for the runs with the paper recipe. Where the text and the code disagree (batch size, biases), I follow the code.

4.4 Receptive-field analysis

The RF of a channel is the gradient of its activation at the image centre with respect to the input pixels. The *black probe* evaluates it at an all-zero image, as the authors' code does; a channel whose gradient is zero there (a ReLU that is off at a black image) is dropped from the averages, as in the code. The *natural probe* averages the gradient over 256 test images, which keeps every channel that responds to real images. The OSI is a port of the authors' code; on 1 000 Gaussian-noise RFs it gives 1.925 ± 0.037 (paper: 1.97 ± 0.08), and this value, computed before the runs, is the noise level of the decision rule. As a check that does not depend on the OSI, every retinal RF is also fitted with an isotropic difference of Gaussians (DoG) and with a Gabor function, and labelled *center-surround* (my OSI < 2 and a DoG fit with $R^2 \geq 0.5$ not worse than the Gabor by more than 0.05), *oriented* (OSI ≥ 2, Gabor $R^2 \geq 0.5$ and better than the DoG), *dead* or *other*;

[1] https://github.com/ganguli-lab/RetinalResources, commit 3235142; files `TrainModel.py`, `VisualizeRFs.py`, `Quantify_Orientedness.ipynb`.

Data	N_{BN}	D	Test acc.	Retina, black	Retina, natural	C-S / or.	Layer 1, black
CIFAR-10	1	2	0.676 ± 0.020	1.83 ± 0.38 (3/3)	1.38 ± 0.13 (3/3)	3 / 0	3.22 ± 0.62 (26/96)
CIFAR-10	4	2	0.684 ± 0.031	2.03 ± 0.38 (8/12)	2.27 ± 0.59 (12/12)	3 / 7	2.93 ± 0.48 (44/96)
CIFAR-10	32	2	0.705 ± 0.018	2.84 ± 0.58 (25/96)	3.29 ± 0.36 (96/96)	0 / 56	2.48 ± 0.33 (48/96)
CIFAR-10	1	4	0.692 ± 0.035	1.58 ± 0.26 (3/3)	1.42 ± 0.40 (3/3)	2 / 0	2.30 ± 0.31 (25/96)
CIFAR-10	4	4	0.708 ± 0.008	1.53 ± 0.17 (9/12)	1.67 ± 0.29 (12/12)	3 / 2	2.22 ± 0.39 (14/96)
CIFAR-10	32	4	0.706 ± 0.017	1.92 ± 0.27 (19/96)	2.40 ± 0.22 (95/96)	2 / 42	2.41 ± 0.26 (31/96)
STL-10	1	2	0.409 ± 0.007	– (0/3)	3.42 ± 1.03 (3/3)	0 / 2	– (0/96)
STL-10	4	2	0.444 ± 0.014	1.31 ± 0.23 (10/12)	3.12 ± 0.97 (12/12)	2 / 9	1.54 ± 0.26 (48/96)
STL-10	32	2	0.508 ± 0.035	1.53 ± 0.28 (50/96)	3.72 ± 0.70 (96/96)	5 / 57	1.34 ± 0.16 (37/96)
Imagenette	1	2	0.473 ± 0.033	2.03 ± 0.78 (3/3)	3.30 ± 1.08 (3/3)	0 / 3	3.23 ± 0.52 (71/96)
Imagenette	4	2	0.520 ± 0.020	4.09 ± 1.57 (10/12)	3.97 ± 2.11 (12/12)	4 / 5	6.44 ± 1.33 (62/96)
Imagenette	32	2	0.595 ± 0.023	2.59 ± 0.33 (59/96)	3.72 ± 0.64 (96/96)	6 / 57	2.34 ± 0.37 (48/96)

Table 3: Code recipe, 3 seeds per row. Retinal and first-cortical-layer OSI: mean $\pm$ 95% interval over channels, with the number of non-zero RFs out of the channels in brackets; the natural probe has no zero RFs except one. C-S / or.: retinal channels labelled center-surround / oriented (natural probe). Paper ($N_{\mathrm{BN}} = 1$, $D_{\mathrm{VVS}} = 2$): retina 1.56 ± 0.22, first cortical layer 3.05 ± 0.30; noise 1.925 (ported OSI).

the thresholds were fixed on synthetic filters before looking at learned ones (40/40 correct per class under 10% noise). Intervals are 95% normal intervals over channels, $\pm 1.96\, s/\sqrt{n}$.

4.5 Experiments

- **Main runs, code recipe (36 networks).** CIFAR-10 control: $N_{\mathrm{BN}} \in \{1,4,32\}$, $D_{\mathrm{VVS}} \in \{2,4\}$, 3 seeds; STL-10 and Imagenette: $N_{\mathrm{BN}} \in \{1,4,32\}$, $D_{\mathrm{VVS}} = 2$, 3 seeds. The differences, the prediction and the rule were committed before the first run; one deviation (matching the number of updates) was decided after three undertrained pilot runs and committed before the rest (Section 5.1).
- **Paper recipe (234 networks).** A sweep of $N_{\mathrm{BN}} \in \{1,\ldots,32\}$ with retina depth 1–2 (72), a cortex-depth grid $D_{\mathrm{VVS}} \in \{1,\ldots,4\}$ at $N_{\mathrm{BN}} \in \{1,4,32\}$ (54), a CIFAR-10 control (18), a recipe check (4), a bias check (2) and a bridge experiment from the control to my setup (72 networks, 5 seeds). Section 6.

5 Results with the code recipe

5.1 Prediction and decision rule

Before running I wrote three rules for the CIFAR-10 control, about the retina at $D_{\mathrm{VVS}} = 2$. **R1** (does the recipe explain the dead channels?): fewer than 30% of the 32-channel retina channels have a zero gradient under the natural probe (91% with the paper recipe). **R2** (the paper's headline): at $N_{\mathrm{BN}} = 1$ the mean OSI of the retina under the black probe is below the noise level 1.925, and that of the first cortical layer is above it. **R3** (the contrast with a wide retina): the retinal OSI is lower at $N_{\mathrm{BN}} = 1$ than at $N_{\mathrm{BN}} = 32$, with non-overlapping 95% intervals. If R1 held, these runs would be the main results and those with the paper recipe secondary. For STL-10 and Imagenette I fixed no pass/fail rule, only the same quantities. *Deviation:* the plan was 20 epochs everywhere, but the first three networks on STL-10 and Imagenette reached a test accuracy of 0.12–0.31: 20 epochs of 4 500 images at batch 64 are 1 420 updates, against 15 640 in the paper. I therefore matched the paper's number of updates on those datasets, which restricted the main runs to the configurations above; the pilot runs are kept.

5.2 The paper's setting (CIFAR-10)

All three rules hold (Table 3). **R1:** no retinal channel of the 32-channel networks is silent under the natural probe (0 of 96, against 91% with the paper recipe). **R2:** the one-channel retina has OSI 1.83 ± 0.38 under the

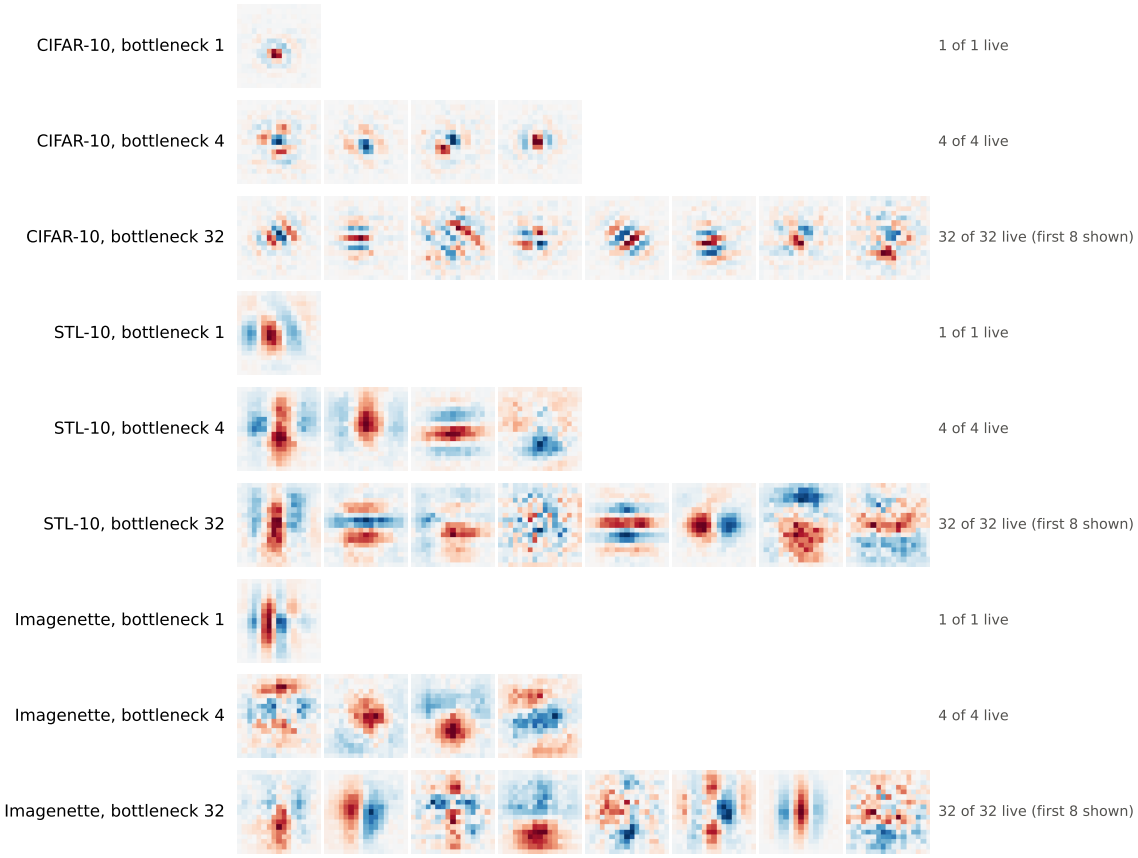


Figure 2: Retinal RFs with the code recipe (seed 0, $D_{\mathrm{VVS}} = 2$, natural probe, central 17×17 pixels; red positive, blue negative, each scaled to its own maximum). On CIFAR-10 the one-channel retina is a center-surround blob and the 32-channel retina is mostly oriented; on STL-10 and Imagenette the one-channel retina is elongated.

black probe (paper 1.56 ± 0.22) and 1.38 ± 0.13 under the natural probe, and all three channels are labelled center-surround; the first cortical layer has 3.22 ± 0.62 (paper 3.05 ± 0.30). **R3:** 1.83 ± 0.38 at one channel against 2.84 ± 0.58 at 32 channels, where 56 of 96 channels are oriented and none is center-surround. R3 holds narrowly: with three one-channel networks the interval is wide, and with the sample instead of the population standard deviation it would overlap (2.30 against 2.24); under the natural probe the gap is wide (1.38 ± 0.13 against 3.29 ± 0.36). Under the black probe 74% of the 32-channel retina channels have a zero gradient, as the authors' code anticipates ("RFs may get stuck with zero gradient due to ReLU nonlinearities"); those channels respond to natural images, so they are silent at a black image but not dead. Fig. 2 shows the RFs: a compact isotropic blob at one channel, elongated filters at 32. Test accuracy is 0.68–0.71, at the top of the range of the paper's Fig. 1D, and grows with the width ($0.676 \rightarrow 0.705$).

Two descriptive results go beyond the rules. With $D_{\mathrm{VVS}} = 4$ the retina is less oriented than with $D_{\mathrm{VVS}} = 2$ at every width (black probe: 1.58, 1.53, 1.92 against 1.83, 2.03, 2.84), in the direction of H1 and H2 but not of the paper, which reports no qualitative dependence on depth; and the accuracy gain from depth 2 to 4 is small ($+0.00$ to $+0.02$, within one standard deviation).

5.3 STL-10 and Imagenette

With the same recipe and the paper's number of updates, *accuracy* grows with the width (STL-10 0.409, 0.444, 0.508; Imagenette 0.473, 0.520, 0.595 for $N_{\mathrm{BN}} = 1, 4, 32$), claim (iii), and all channels respond to natural images. The *retinal geometry* does not follow the paper. Under the natural probe the one-channel retina is oriented on both datasets (OSI 3.42 on STL-10 and 3.30 on Imagenette, no network below the noise level, none of six channels center-surround), as oriented as the 32-channel retina (3.72 on both), and

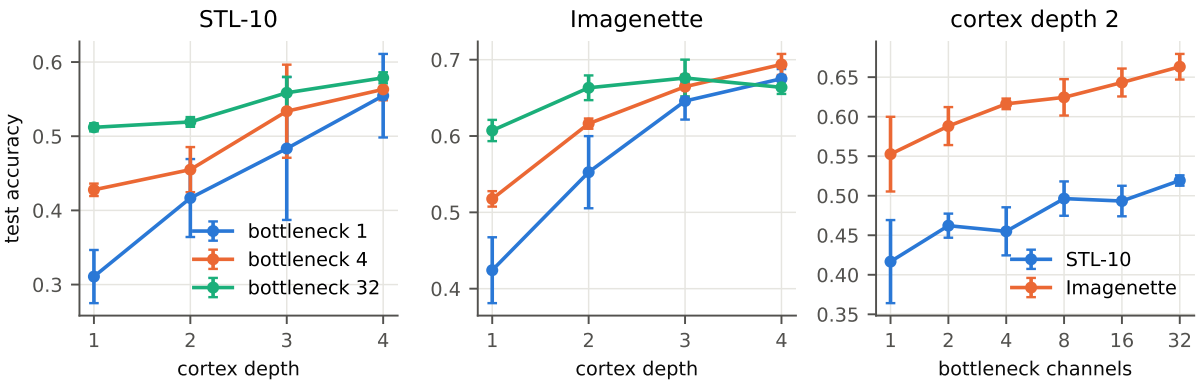


Figure 3: Paper recipe: test accuracy (mean ± s.d. over 3 seeds) against cortex depth for three bottlenecks (left, centre) and against bottleneck width at cortex depth 2 (right).

the RFs are elongated (Fig. 2). The black probe does not rescue the claim: on STL-10 the one-channel retina has a zero gradient at a black image in all three networks, and the first cortical layer is itself below the noise level (1.34–1.54), so the probe does not separate retina from cortex at this resolution; on Imagenette it gives 2.03, 4.09 and 2.59 for $N_{\mathrm{BN}} = 1, 4, 32$, with no trend. The first cortical layer is oriented under the natural probe (median 7.0 and 5.2 at one channel), but so is the retina. Retinal linearity is highest at one channel (0.73 on STL-10 and 0.76 on Imagenette, against 0.66 and 0.69 at 32 channels), as in the paper, but it is not monotonic on Imagenette (0.62 at 4 channels); the information kept by the retina grows with the width (0.10 → 0.87 and 0.23 → 0.96).

6 Results with the paper recipe

These runs used lr 10^{-3}, batch 32, centred inputs, flip only and the best-validation checkpoint (Table 2), the natural probe and my own OSI on a 17×17 crop (on noise: mean 1.77, median 1.67, 74% below 2, so the threshold 2 of the labels is at noise level). They cover the full sweep and the cortex-depth grid, which were not repeated with the code recipe.

Accuracy (claim iii). Test accuracy increases with the cortex depth in all six dataset–bottleneck cells (Fig. 3): at $N_{\mathrm{BN}} = 1$ from 0.31 to 0.56 on STL-10 and from 0.42 to 0.68 on Imagenette between depth 1 and 4. At depth 2 it increases with the width, from 0.417 to 0.519 (STL-10) and from 0.553 to 0.663 (Imagenette) between 1 and 32 channels, with no knee, so H3 (little loss until a minimum width) is not confirmed. 12 of 126 runs collapsed at the first attempt and were retrained with lr/3.

Retinal RFs (claim i). Of the 702 retinal channels of the sweep, 22 are labelled center-surround, 270 oriented, 316 other and 94 dead (13%); the median OSI of the live channels is 2.5, and 37% are below 2 but only 12% below 1.40 (the 25th percentile of noise). At $N_{\mathrm{BN}} = 1$, $D_{\mathrm{VVS}} = 2$ the one-channel retina has OSI 1.22, 1.51, 1.69 on Imagenette (mean 1.47) and 1.84, 3.23, 3.62 on STL-10 (mean 2.90). In the configuration of H1 ($N_{\mathrm{BN}} = 2$, $D_{\mathrm{VVS}} = 4$) 1 of 6 channels is center-surround on each dataset. At fixed width, depth 2 has more oriented channels than depth 4 in three of four comparisons (e.g. $N_{\mathrm{BN}} = 16$: 22/48 against 14/48 on STL-10), as H2 is worded. Table 4 gives the details. The first cortical layer (Fig. 4) has median OSI 2.4–5.3 and is more oriented than the retina at $N_{\mathrm{BN}} \leq 4$ in 4 of 6 cells.

Linearity and information (claim ii). Retinal linearity decreases with the width at depth 2 (Imagenette 0.73 → 0.42, STL-10 0.62 → 0.39 from 1 to 32 channels) and increases with the cortex depth only at $N_{\mathrm{BN}} = 1$ (0.52 → 0.76 on STL-10, 0.65 → 0.76 on Imagenette, depth 1 to 4), the case the paper highlights;

Dataset	N_{BN}	Channels	C-S	Orient.	Other	Dead	Median OSI	OSI<2
STL-10	1	3	0	2	1	0	3.23	1/3
STL-10	2	6	1	5	0	0	2.33	1/6
STL-10	4	12	0	6	5	1	2.71	5/11
STL-10	8	24	0	15	4	5	3.26	2/19
STL-10	16	48	1	22	15	10	2.63	14/38
STL-10	32	96	2	49	22	23	2.85	23/73
Imagenette	1	3	1	0	2	0	1.51	3/3
Imagenette	2	6	3	3	0	0	2.33	3/6
Imagenette	4	12	0	10	1	1	2.87	0/11
Imagenette	8	24	1	13	7	3	3.51	6/21
Imagenette	16	48	2	28	13	5	2.86	14/43
Imagenette	32	96	2	54	26	14	3.56	18/82

Table 4: Paper recipe: labels of the retinal channels of the sweep (retina depth 2, cortex depth 2; C-S = center-surround), median OSI of the live channels and number of live channels with OSI < 2.

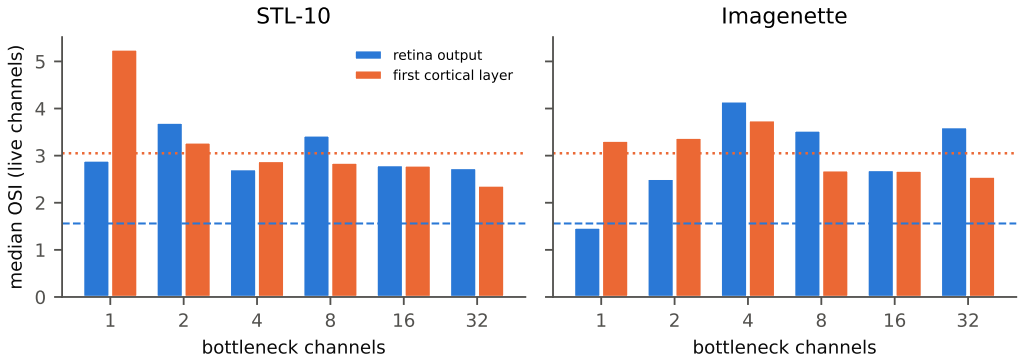


Figure 4: Paper recipe: median OSI (my implementation) of the live channels of the retinal output and of the first cortical layer (cortex depth 2). Dashed: paper's retina (1.56); dotted: paper's first cortical layer (3.05).

at 4 and 32 channels it is flat or peaks at depth 2 (Fig. 5). The information kept by the retina (local reconstruction of the centre pixel from a 9×9 window of retinal output) increases with depth in 4 of 6 cells.

CIFAR-10 control and why it failed. With the paper recipe (Table 5) the paper's own setting gave accuracy 0.56–0.67, 6 of 18 collapsed first attempts and 67% of the retinal channels dead under the natural probe; the one-channel retina still had OSI 1.60 (three seeds), but its pre-registered prediction (at depth 4) missed both bounds narrowly. A recipe check (input centring, flip, checkpoint; one seed) changed nothing, and initialising the biases from Glorot-uniform, as the text says, left 4 and 13 of 32 channels alive (two variants; zero biases: 3). The dead channels are dying ReLUs: none is dead at initialisation, and in one 32-channel network 16 of the 29 channels silent at the image centre never fire on any test image (their weights shrink to a third of the norm of the live ones), 6 fire only near the border and 7 on at most 1.2% of the images. The learning rate of 10^{-3}, ten times the authors', is the classical cause, and the runs with the code recipe confirm it (R1).

Bridge experiment. With the same recipe I also walked from the failed control to my setup one factor at a time (fewer images, STL-10 images, pooling head, 96 pixels; 5 seeds; Table 6). By its registered rule (contrast Δ between the median OSI at $N_{\mathrm{BN}} = 32$ and at $N_{\mathrm{BN}} \in \{1,4\}$) it was inconclusive, because the control itself had 145 of 160 channels dead at $N_{\mathrm{BN}} = 32$; the design was not blind to the earlier results. Post hoc, the one-channel retina had OSI < 2 in 15 of 20 networks with the paper's head or CIFAR-10 images, and in 3 of 10 with STL-10 images and my pooling head ($p = 0.045$, uncorrected; $p = 0.63$ with the threshold 1.23); the dead channels followed the head (66–91% with flatten and dense layers, 9–29%

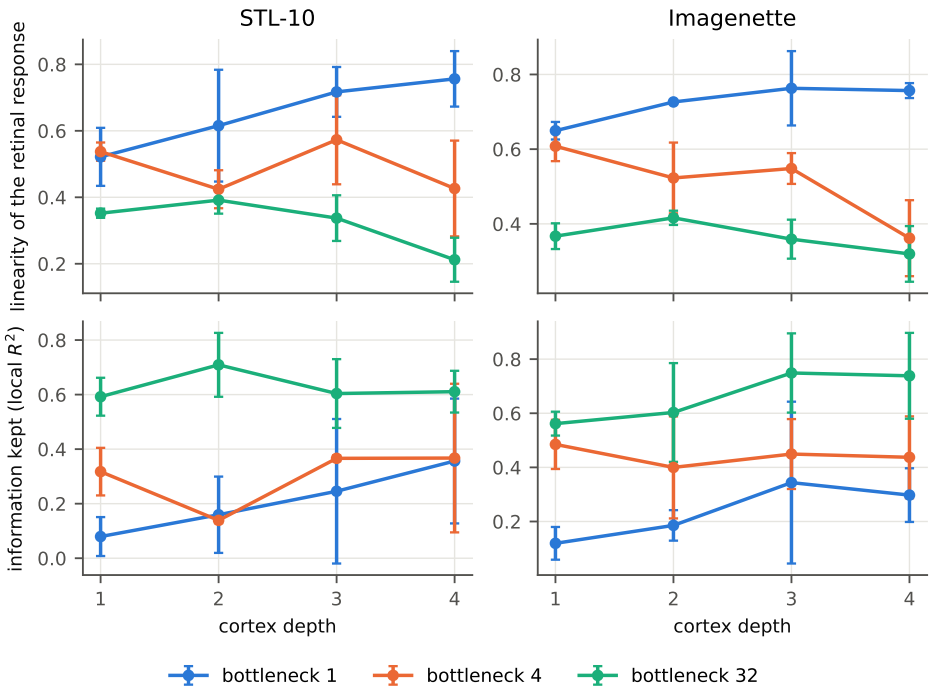


Figure 5: Paper recipe: linearity of the retinal response and local information kept by the retina against cortex depth (mean ± s.d. over 3 seeds, live channels only).

N_{BN}	Depth	Test acc.	Collapsed	Live/ch.	C-S	Median OSI	OSI<2
1	2	0.672 ± 0.004	3	3/3	1	1.25	2/3
1	4	0.652 ± 0.079	2	3/3	0	2.31	1/3
4	2	0.605 ± 0.024	0	4/12	2	1.58	3/4
4	4	0.555 ± 0.022	0	4/12	2	1.85	2/4
32	2	0.594 ± 0.021	0	9/96	1	2.44	4/9
32	4	0.573 ± 0.092	1	50/96	1	2.55	18/50

Table 5: Paper recipe, CIFAR-10 control (paper's architecture, natural probe): collapsed first attempts, live channels out of the channels of the 3 seeds, center-surround labels and OSI (my implementation).

with pooling). With this recipe these numbers mix the effect of the setting with that of dying ReLUs, so I do not use them for conclusions.

Condition	Images	Train	Res.	Head	Dead, N_{BN}=32	OSI, N_{BN}=1 (<2)	Δ [90% interval]
control	CIFAR-10	45 000	32	flatten + FC	145/160	1.25 (4/5)	0.24 [−0.66, 1.20]
data	CIFAR-10	4 500	32	flatten + FC	124/160	1.48 (3/5)	−0.05 [−0.58, 0.63]
stl32	STL-10	4 500	32	flatten + FC	106/160	1.41 (4/5)	0.52 [0.24, 0.65]
stl32_pool	STL-10	4 500	32	pool + GAP	26/160	2.69 (1/5)	−0.05 [−0.37, 0.31]
sweep	STL-10	4 500	96	pool + GAP	46/160	2.95 (2/5)	0.27 [−0.33, 0.75]
head	CIFAR-10	45 000	32	pool + GAP	14/160	1.36 (4/5)	0.44 [−0.13, 0.73]

Table 6: Paper recipe, bridge experiment (retina depth 2, cortex depth 2, 5 seeds): channels dead out of 160 at $N_{\mathrm{BN}} = 32$, median OSI of the five one-channel retinas (in brackets: how many below 2), and the registered contrast Δ with its 90% bootstrap interval.

7 Discussion

The paper's claim holds in its own setting. With the code recipe, three seeds and my port of their analysis, CIFAR-10 gives back the paper's numbers: an isotropic, center-surround one-channel retina (1.83 ± 0.38 against 1.56 ± 0.22), an oriented first cortical layer (3.22 ± 0.62 against 3.05 ± 0.30) and an oriented 32-channel retina. The registered contrast with the wide retina holds, narrowly on the black probe and clearly on the natural one. Table 7 summarises each claim.

It does not transfer to the assignment's datasets. On STL-10 and Imagenette, with the same recipe, matched updates and all channels alive, the one-channel retina is as oriented as the 32-channel one. This rules out the explanations that the paper recipe left open: dead channels, an unsuitable learning rate, undertraining and my analysis (which returns the paper's numbers on CIFAR-10). What remains is the setting: the images (natural photographs at 96×96, where a 9×9 kernel covers a smaller part of the object than at 32×32), the smaller training sets and the pooling head, which change together between the two settings. I did not run the bridge experiment with the code recipe, so I cannot say which of the three matters; the bridge experiment with the paper recipe pointed at the combination of STL-10 images and the pooling head, but with a recipe that kills channels. The answer to the question "is it the datasets or the resources?" is therefore: the setting, not the compute, since the paper's setting is reproduced with the same compute and number of seeds.

Paper against code. The text and the code disagree on the batch size (32 against 64) and the biases (Glorot against zero), but the decisive difference is one the text leaves open: the learning rate. The Keras default I used for it (10^{-3}, ten times the code's) gives dying ReLUs (dead channels and collapsed runs) that blur every width comparison; the Glorot biases of the text do not change this (Section 6). The paper's result needs the code recipe. The runs with the paper recipe still supply the only evidence on the cortex depth: accuracy grows with depth, and linearity grows with depth only at one channel.

Relation to the assignment. H1 (narrow bottleneck and deep cortex give center-surround filters): in the paper's setting, 2 of 3 one-channel retinas and 3 of 12 four-channel channels at $D_{\mathrm{VVS}} = 4$ are center-surround, and the retina is less oriented at depth 4; on STL-10 and Imagenette not tested at depth 4 with the code recipe, and not reproduced with the paper recipe. H2 (wide bottleneck or shallow cortex give oriented filters): supported in the paper's setting (56 of 96 oriented channels at 32 channels and depth 2), and for the width trivially true on STL-10 and Imagenette, where every width is oriented. H3: not confirmed as a threshold; accuracy decreases gradually as the bottleneck narrows.

Claim of the paper	CIFAR-10, code recipe	STL-10 and Imagenette
(i-a) Narrow bottleneck: isotropic, center-surround retina	**Reproduced**: 1.83 ± 0.38 (black), 1.38 (natural), 3/3 C-S	**Not reproduced**: OSI 3.42, 3.30 (natural), 0/6 C-S; black probe uninformative
(i-b) Wide bottleneck: oriented retina	**Reproduced**: 2.84 (black), 56/96 oriented	Oriented (3.72), but so is the narrow retina
(i-c) Oriented first cortical layer	**Reproduced**: 3.22 ± 0.62	Natural probe: yes (7.0, 5.2 at $N_{\mathrm{BN}} = 1$); black probe on STL-10: no
(ii-a) Linearity decreases with width	not measured	Partly: $0.73 \to 0.66$, $0.76 \to 0.69$ (1 to 32 channels), dip at 4 on Imagenette
(ii-b,c) Deeper cortex: more linear, more information	not measured	paper recipe only: at $N_{\mathrm{BN}} = 1$ only; information in 4 of 6 cells
(iii-a) Accuracy grows with depth	$+0.00$ to $+0.02$ ($2 \to 4$), within noise	paper recipe: 6 of 6 cells
(iii-b) Accuracy grows with width	**Reproduced**: $0.676 \to 0.705$	**Reproduced**: $0.41 \to 0.51$, $0.47 \to 0.60$

Table 7: The claims of the paper against my results. Main runs unless marked "paper recipe".

8 Limitations

- Three seeds instead of ten; one channel per network at $N_{\mathrm{BN}} = 1$, so the headline numbers rest on three RFs per cell, and R3 holds only narrowly on the black probe.
- The main runs cover $N_{\mathrm{BN}} \in \{1, 4, 32\}$ at $D_{\mathrm{VVS}} = 2$ on STL-10 and Imagenette (one of three seeds of $N_{\mathrm{BN}} = 4$ on STL-10 had to be rerun after a CUDA error); the cortex-depth claims rely on the paper recipe, which kills channels. Linearity was not measured on CIFAR-10.
- The matched number of updates is a deviation from the code (which uses 20 epochs), decided after three pilot runs; the head at 96×96 differs from the paper's, and data amount, image content, resolution and head are confounded in the comparison between CIFAR-10 and the assignment's datasets.
- The black probe is uninformative on STL-10 (even the first cortical layer is isotropic under it), so the conclusion there rests on the natural probe, which the paper does not use. The center-surround labels depend on thresholds set on synthetic filters.
- The bridge experiment uses the paper recipe; its design was not blind to earlier results and its one-channel pattern is post hoc.

9 Conclusions

With the training recipe of the authors' code, which differs from the text of the paper, the central result of Lindsey et al. is reproduced in the paper's own setting: on grayscale CIFAR-10 a one-channel retinal bottleneck yields an isotropic, center-surround retina (1.83 ± 0.38, paper 1.56 ± 0.22) followed by an oriented first cortical layer (3.22 ± 0.62, paper 3.05 ± 0.30), while a 32-channel retina is oriented. On the assignment's datasets, STL-10 and Imagenette at 96×96, the same recipe with the paper's number of updates reproduces the dependence of accuracy on the bottleneck width, but not the center-surround retina: the one-channel retina is oriented. Since the analysis and the recipe work on CIFAR-10, the difference lies in the setting (image content and resolution, amount of data, classifier head), which this project does not separate. The paper recipe, with the Keras default learning rate, kills most retinal channels, and with it the claims about cortex depth hold only partly in my setting.

A Reproducibility

Hardware and time: one laptop GPU (NVIDIA RTX PRO 2000 Blackwell, 8 GB). Main runs about 8.5 hours overnight with up to seven networks in parallel; paper recipe about 10 hours. All code is

in `retina_bottleneck.ipynb` (the long training cells were run detached by `runs/nbexec.py`); main results are in `results/official/` (the 20-epoch pilot runs in `pilot_20epochs/`), results with the paper recipe in `results/`; figures and tables are generated by `report/make_figures.py`. The decision rule of the main runs and the deviation are in the git history before the results. A first analysis of the runs with the paper recipe probed the retina without the input centring of the forward pass; it was corrected (center-surround channels 11 $\rightarrow$ 22 of 702) and the old outputs are kept in `results/*_rawprobe.csv`.

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